I have no title, yet.

Git for Scientists

Lars Schöbitz
Global Health Engineering - ETH Zurich

May 27, 2025

https://gitforsci-cis.github.io/website/

Welcome!

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Meet the lecturer

Lars Schöbitz

- Environmental Engineer 🔨
- Retired researcher 🛌
- RStudio certified instructor
- Data steward at ETHZ

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Learning objectives

By the end of this workshop, participants will be able to:

1. Create and clone repositories, and explain the difference between cloning and forking a repository.
2. Apply the standard Git workflow using RStudio IDE—including pull, stage, commit, and push changes to GitHub.
3. Work with branches and pull requests by creating and managing development branches (using both RStudio and the terminal), submitting pull requests, and reviewing and commenting on others' pull requests to facilitate code review and collaborative development.

Teaching strategy

- Teaching Git on a “need to know” basis
- Teaching only a very small subset of Git’s functionality
- Creating a safe space for learning and getting support (use Element)
- Focus on collaboration aspects of GitHub

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Schedule

Time	Module
10:00 - 10:15	GitHub for Collaboration
10:15 - 10:45	Create & Clone
10:45 - 11:00	Break
11:00 - 11:15	Clone & Branch
11:15 - 11:30	Pull Request & Merge
11:30 - 11:35	Publish
11:35 - 11:45	Wrap-up

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Your turn

Think about the last time you published a written document:

- Which tasks gave you joy?
- Which tasks were challenging or frustrating?

Take some written notes.

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02:00

GitHub for Collaboration?

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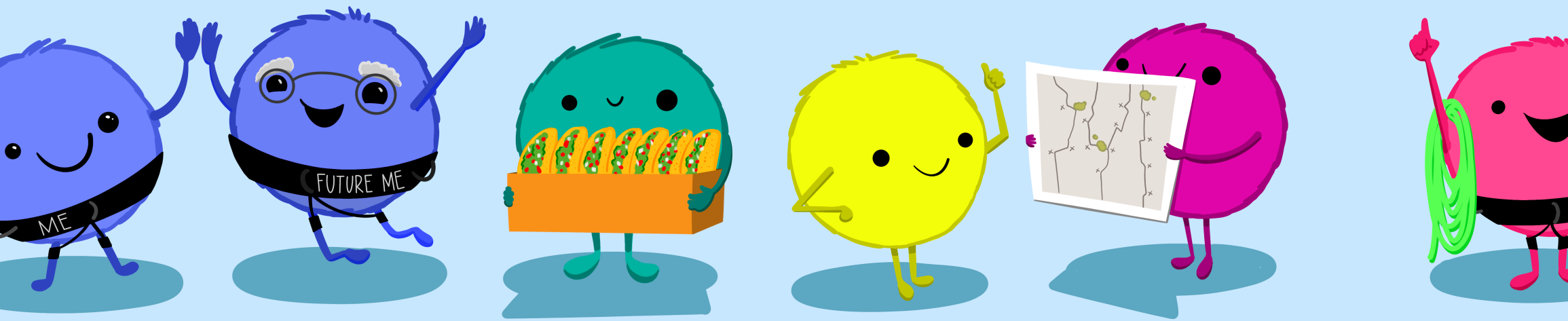
"**Collaboration is the most compelling reason to manage a project with Git and GitHub**. My definition of collaboration includes hands-on participation by multiple people, including your past and future self, as well as an asymmetric model, in which some people are active makers and others only read or review."

-JENNY BRYAN



Bryan, J. 2017. Excuse me, do you have a moment to talk about version control? PeerJ Preprints. 5:e3159v2. DOI: 10.7287/peerj.preprints.3159v2

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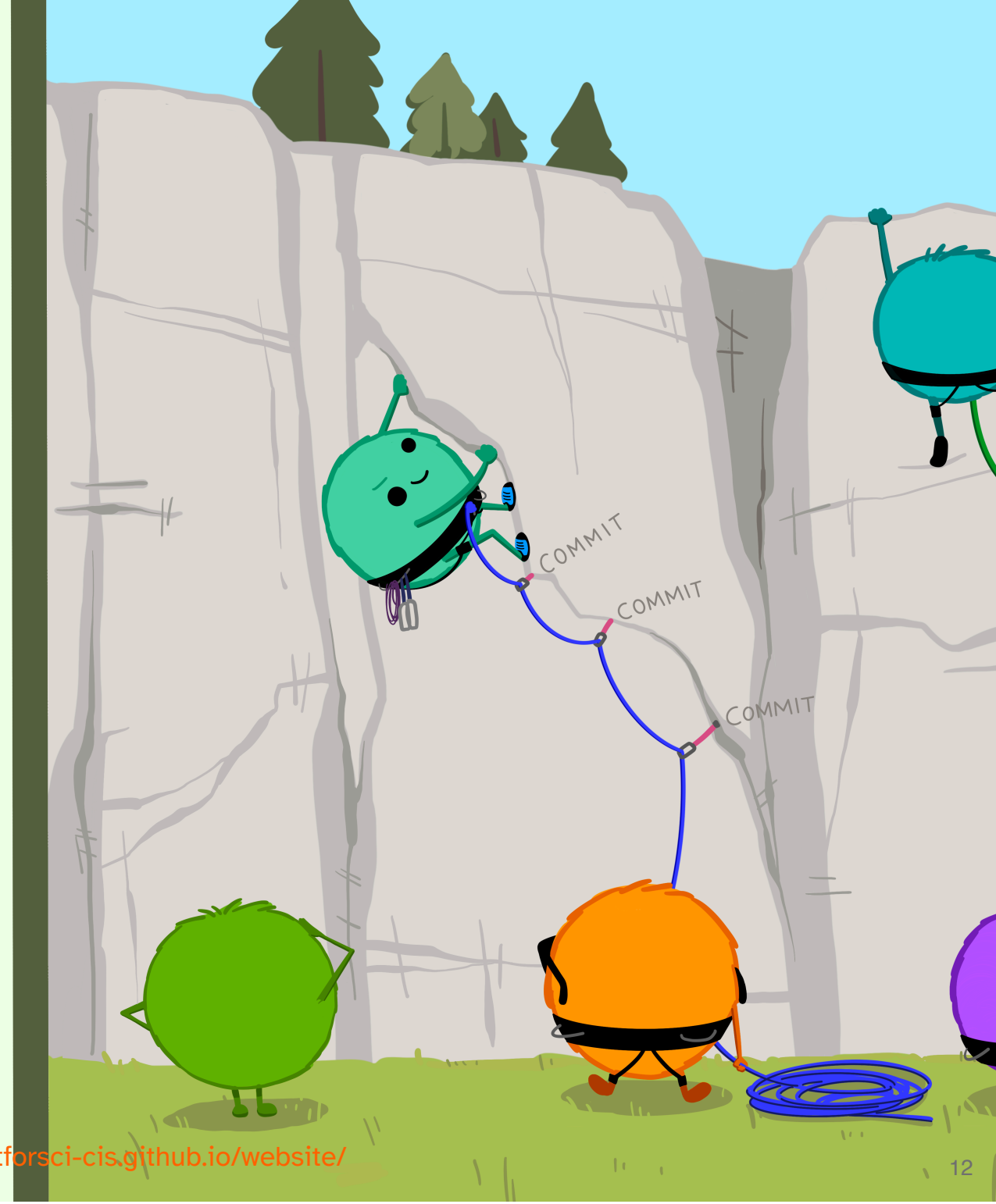
ing a Git commit is like using
chors and other protection when
mbing…**if you make a mistake,
u can't fall past the previous
mmit**.

mmits are also helpful to
ners, because **they show your
urney, not just the destination**.

– HADLEY WICKHAM & JENNY BRYAN



kham & Bryan, RPackages (https://r-packages.org/preface.html)

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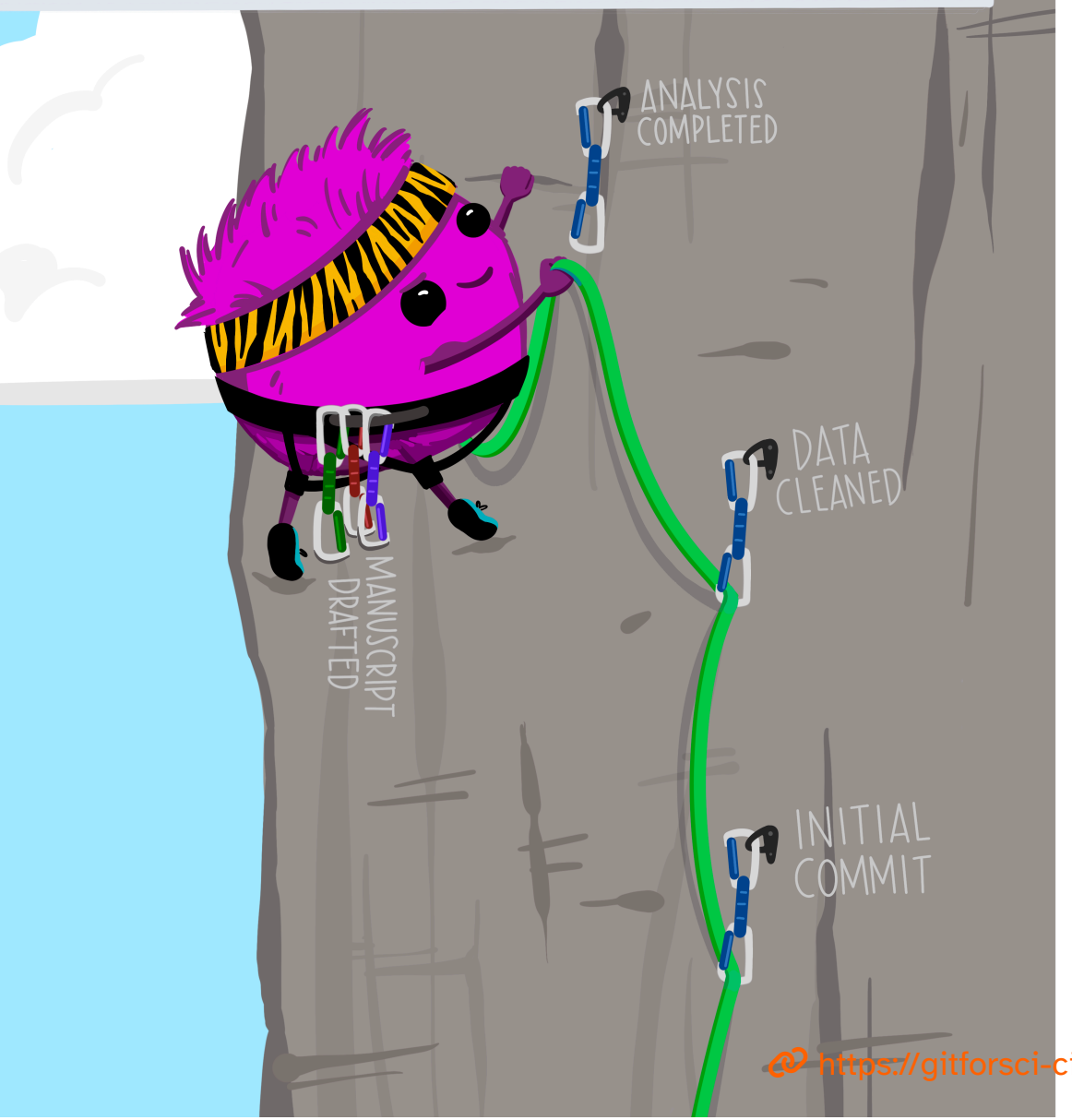
n working with GitHub, we can navigate with
e obvious, safe, streamlined routes that let us
cus on the science-y things we want to do...



...but working without GitHub can be disorien
with too much time spent sifting through p
work to figure out next steps forward.

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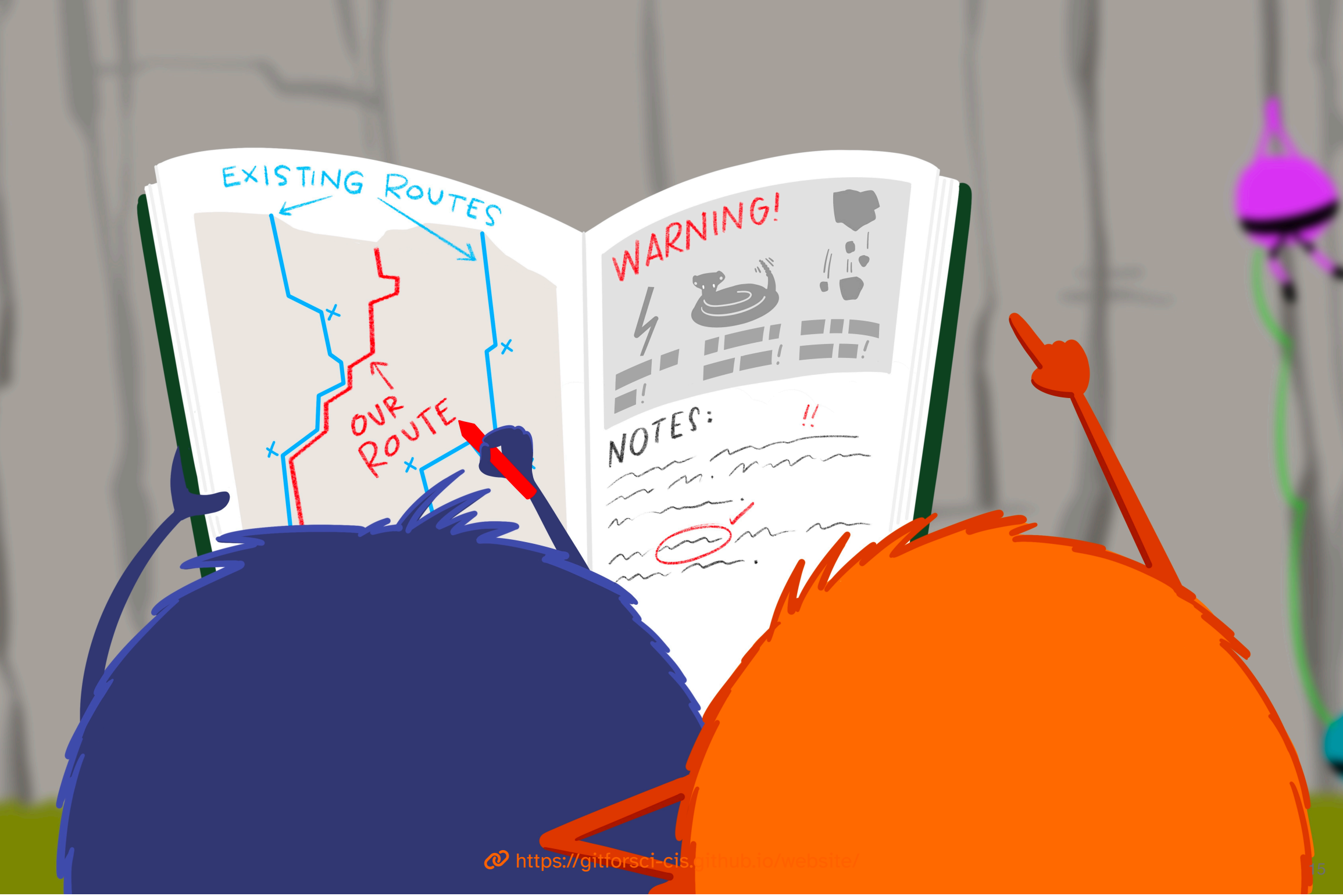
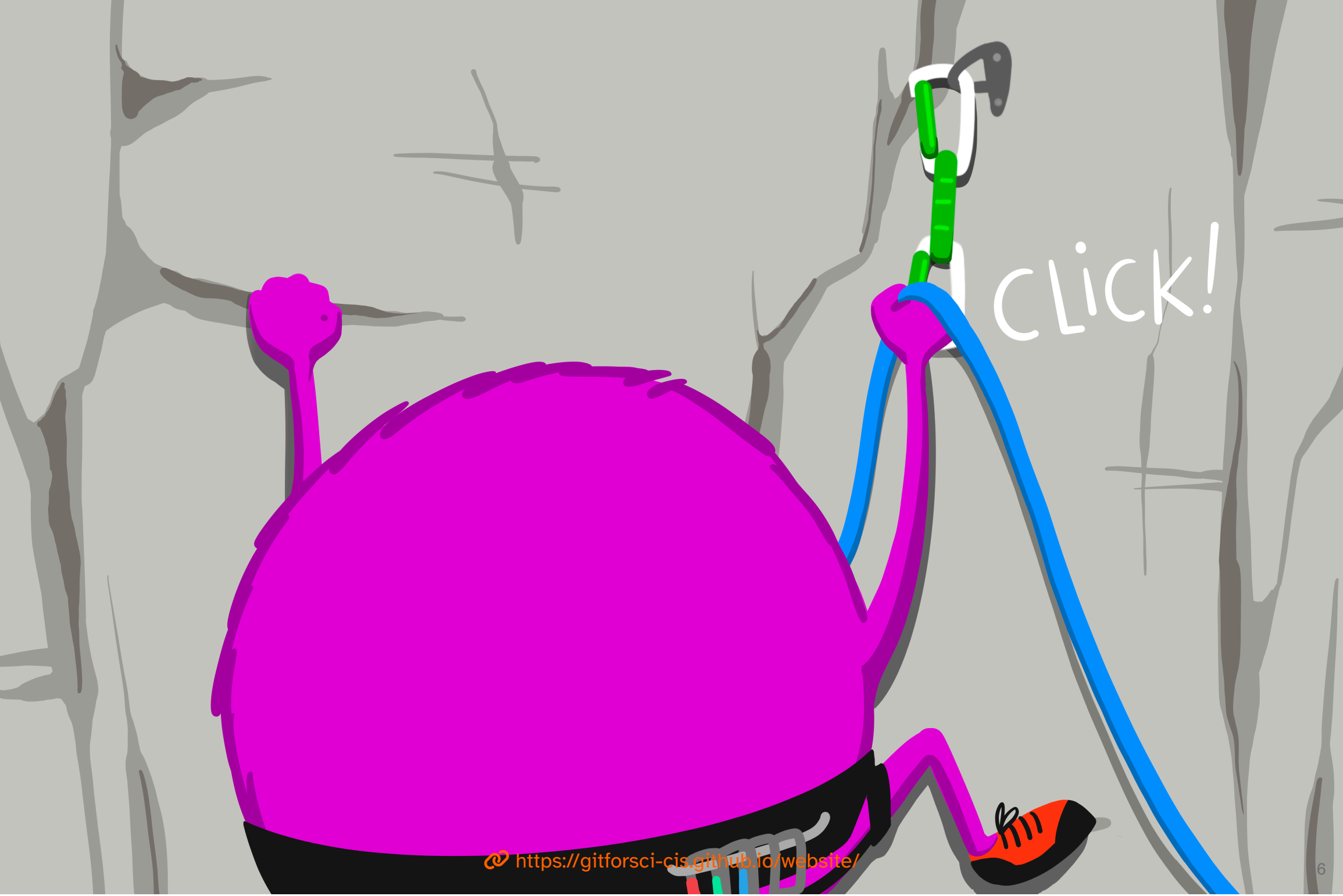

EXISTING ROUTES
OUR ROUTE
WARNING!
NOTES:
!!



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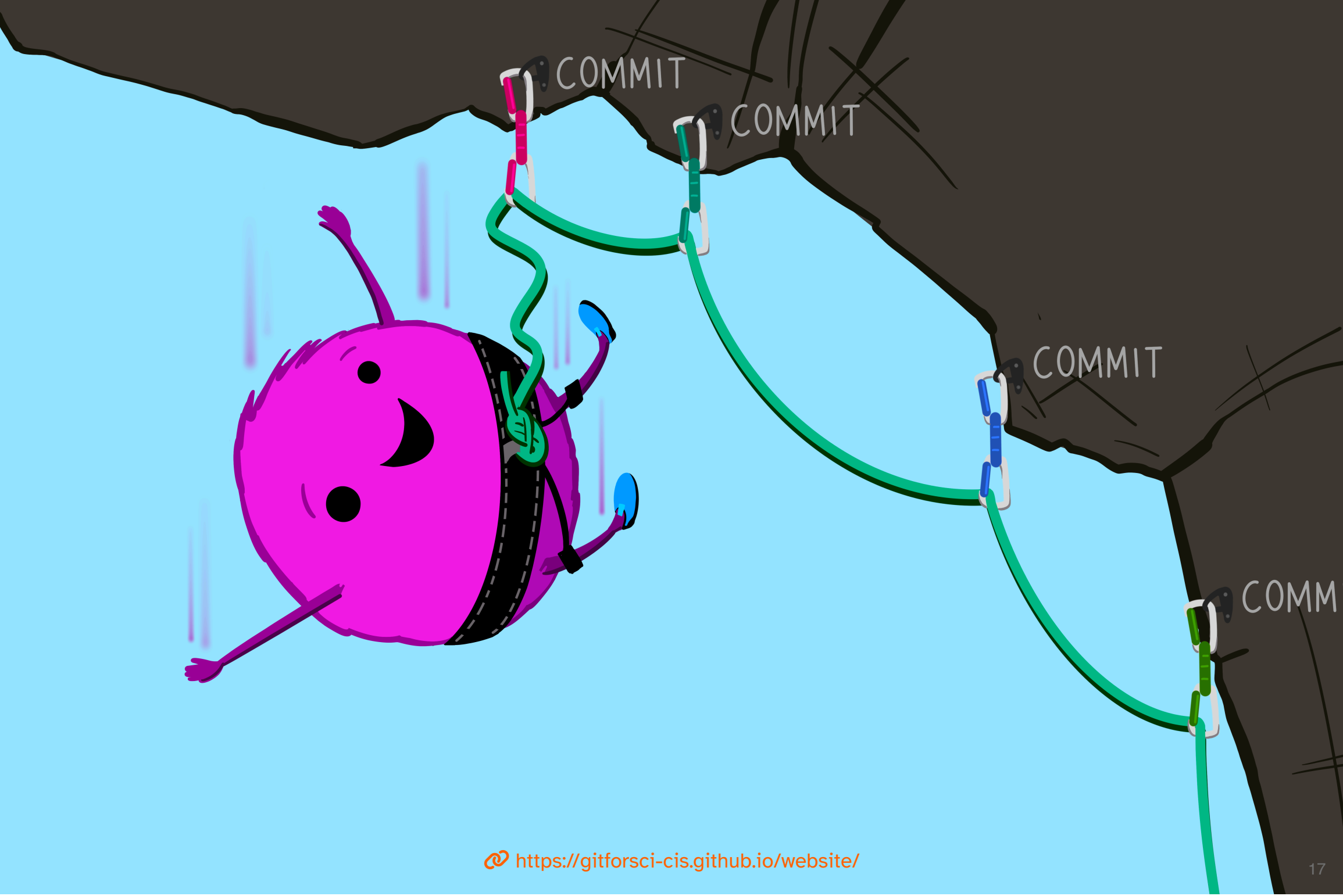

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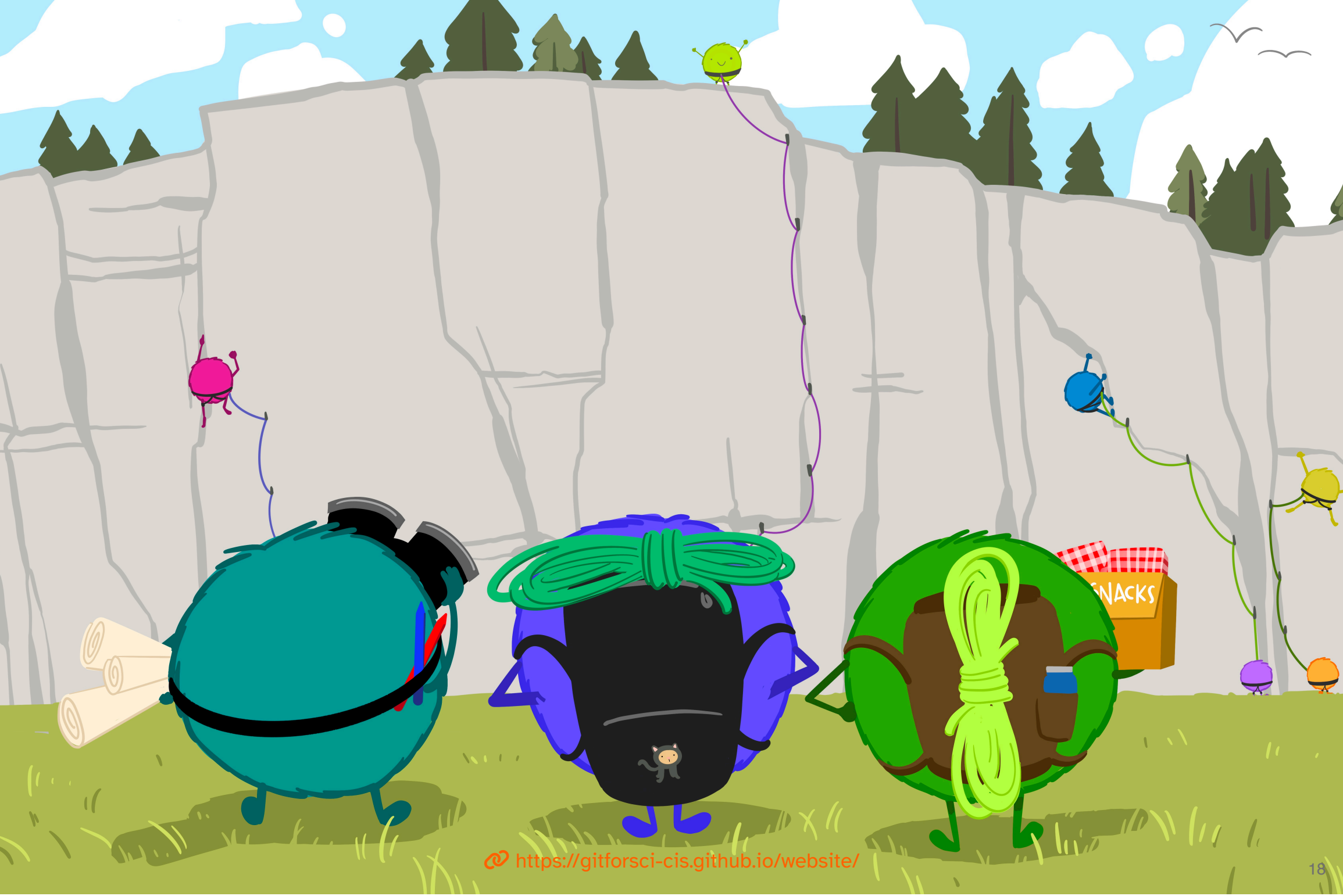

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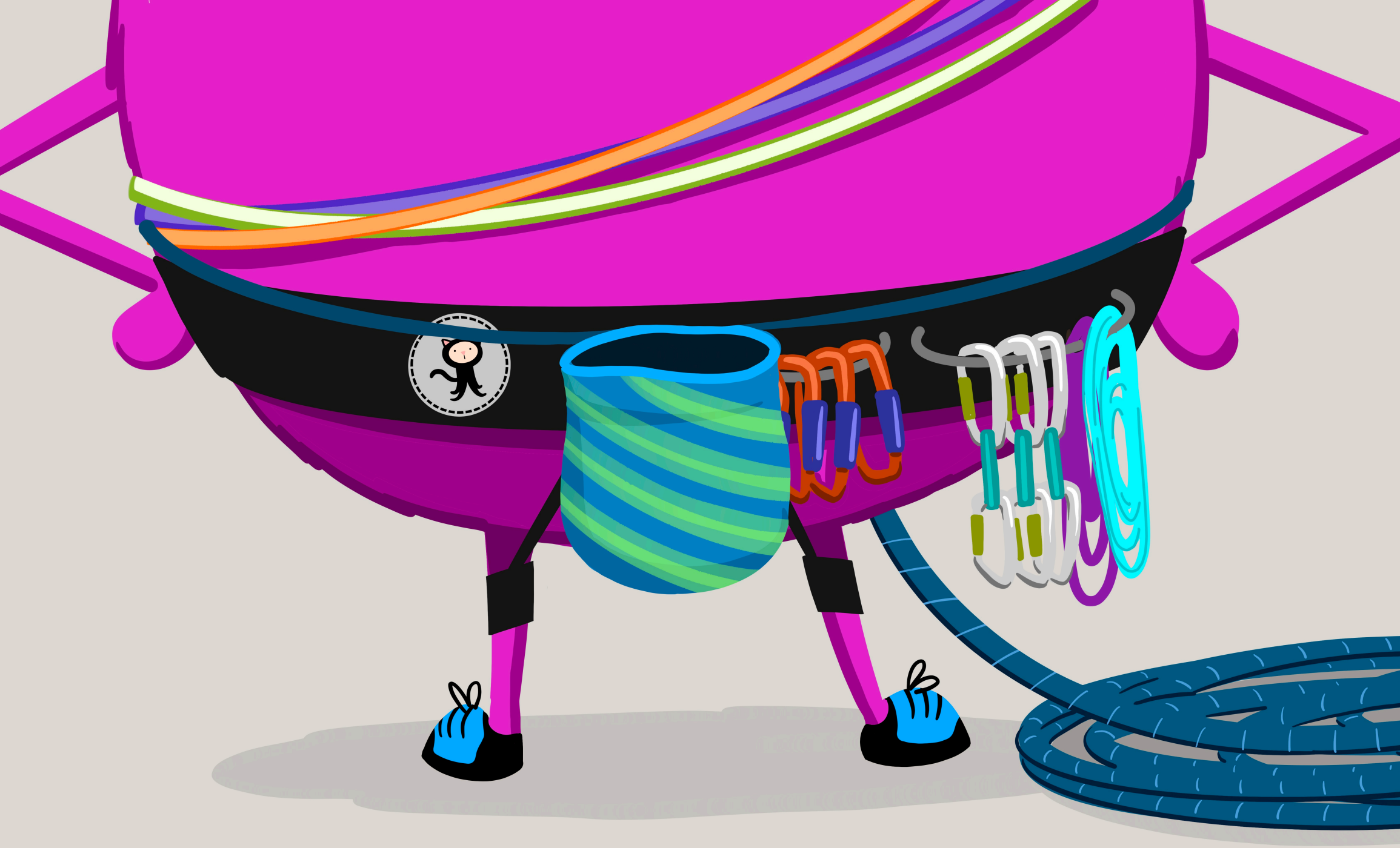

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Read on: GitHub strategies

1. Openscapes Champions Lesson Series. Lowndes, Butland, and Robinson (2024): https://openscapes.github.io/series/core-lessons/github/
2. Excuse Me, Do You Have a Moment to Talk About Version Control? https://peerj.com/preprints/3159/

Create & Clone

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Your turn

- Open github.com
- Navigate to your profile
- When completed, place a yellow sticky note on your laptop

Now, follow along the screen and instructor.

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01:00

Your turn

1. Clone a GitHub repository of a fellow student (browse: github.com/username to fine the repository `website`)
2. Open the repository in RStudio IDE
3. Make a change the `README.md` file
4. Save the file
5. Stage, Commit & Push the changes

id	username1	username2
1	Madliboom	fdaguanno
2	TillRoost	mag-wer
3	pacotvj99	louma3000
4	ellahenninger	CamilleFournierDL
5	dheimgartner	jineichen
6	vgeier	rjalcock
7	Jonathan-Elkobi	rainbow-train

what happens?

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05:00

Read on: Fork and clone

- Happy Git and GitHub for the useR

Take a break

Please get up and move! Leave your emails alone.

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15:00

Clone & Branch

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Your turn

1. Open the landing page for the GitHub organization 'gitforsci-cis': https://github.com/gitforsci-cis (try to use your bookmark)
2. Navigate down the page to the section "Repositories" and use the search field to find the repository called `man-washinvestments-USERNAME` where USERNAME is your GitHub username.
3. Click on the repository name to open it.
4. Click on the green "Code" button.
5. Copy the HTTPS URL to your clipboard by clicking on the clipboard sign.
6. Open the RStudio IDE on your laptop.
7. Click on File > New Project in the top menu bar
8. Click on Version Control.
9. Click on Git.
10. Click on Browse... button.
11. Use the folder structure you created in the pre-course work (e.g. `~/Documents/gitrepos/gh-org-gitforsci-cis`)
12. Paste the HTTPS URL from GitHub into the "Repository URL" field.
13. Click on the "Create Project" button.
14. Find the Files tab in the bottom right pane of RStudio.

Place the yellow sticky note on your laptop when you have completed the task.

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05:00

Pull Request & Merge

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Our turn

- We will continue together with `man-washinvestments-USERNAME` where USERNAME is your GitHub

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01:00

Publish

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Our turn

- We will continue together with `man-washinvestments-USERNAME` where USERNAME is your GitHub

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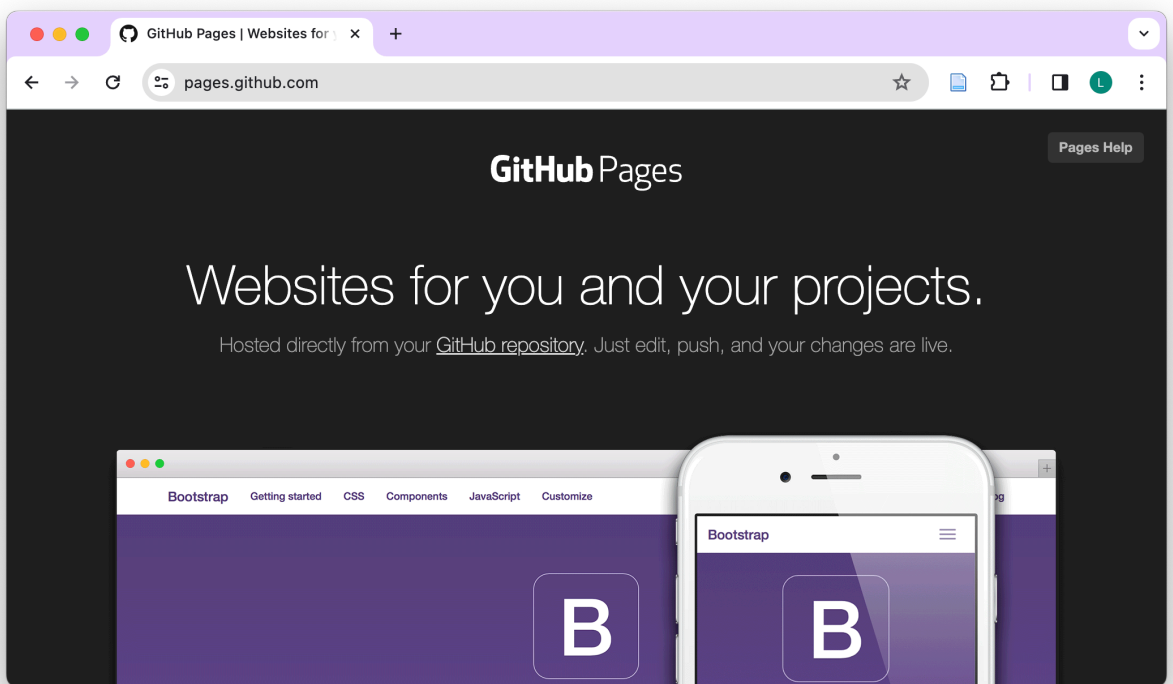
01:00

GitHub Pages

- GitHub Pages is a free service for hosting static websites. It is ideal for blogs, course or project websites, books, presentations, and personal hobby sites.



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Minimal Example - Requirements

- Landing site needs to be stored as `index.qmd`
- The `index.qmd` needs to be stored in `docs` folder
- Example works well for a report/article as a stand-alone page
- Quarto provides a framework and examples for more complex websites: https://quarto.org/docs/websites/

Quarto websites

- Websites are essentially `format: html` + a Quarto Project
- But a website is different than `format: html` in that it has multiple pages
- Websites can be your first exploration into Quarto Projects
- Websites and books are very similar in that they associate multiple pages/resources into a connected resource

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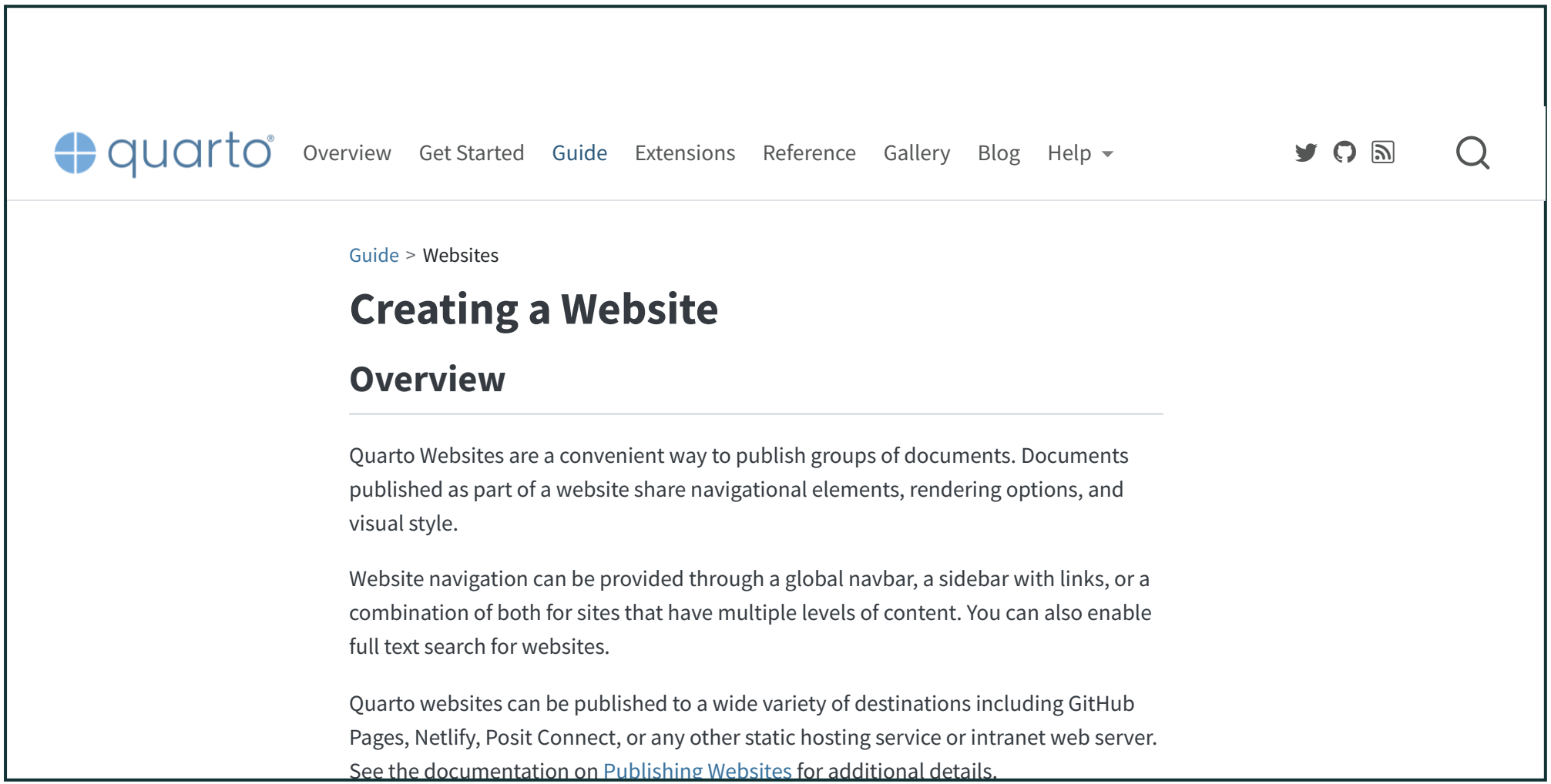
Learn more

quarto.org/docs/websites



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Wrap-up

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Wrap-up

- Post your collected questions in the Element channel
- Start your first project using Git & GitHub (already have a project? Read here)
- Keep learning by doing (create PRs to yourself)
- Share your knowledge with others (help them get started)

Thanks!

Slides created via revealjs and Quarto: https://quarto.org/docs/presentations/revealjs/

Access slides as PDF on GitHub

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References

Lowndes, Julia Stewart, Stefanie Butland, and Erin Robinson. 2024. “Openscapes/Series: Openscapes Lesson Series V2024.11.0.” Zenodo. https://doi.org/10.5281/zenodo.14428503.